Chemistry H Review Guide

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Range: Chapter 5 + 13

Note: (Review Guide/Review Practices/Practice Set) are UNOFFICIAL sources of reference and do NOT include or hint at questions in the actual exam!

Chapter 5 Thermochemistry

Thermochemistry = the study of energy and its transformation in chemical reaction

5.1 The Nature of Energy

- Energy = the ability to do work or transfer heat
- Work = the energy used to cause an object that has mass to move
- Heat = the energy used to cause the temperature of an object to rise

- Work = Force x Distance//

Figure 3(b)

- Heat flows from warmer objects to cooler objects.
- Kinetic energy = the energy an object possesses by virtue of its motion
    - KE = $(mv^2)/2$
- Potential energy = the energy an object possesses by virtue of its position or chemical composition
    - Gravitational PE = mgh

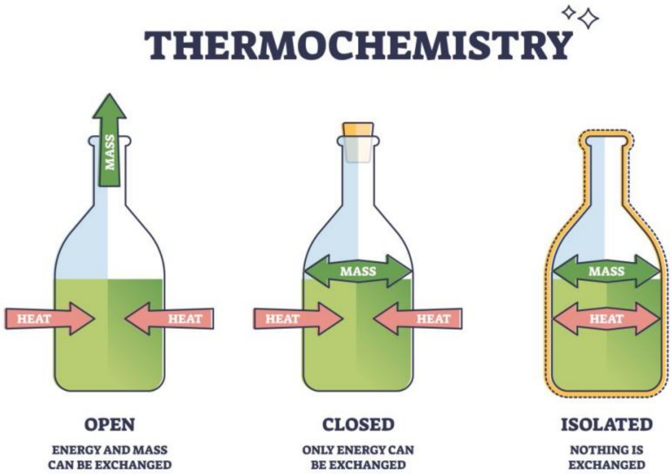
! The lower the energy of a system, the more stable it is

! Forming chemical bonds is exothermic; breaking chemical bonds is endothermic

5.2 The First Law of Thermodynamics

- The First Law of Thermodynamics = energy can be converted from one form to another, but it is neither created nor destroyed
- System = the portion we single out for study

- Surroundings = everything else other than the system



- Internal energy (E) = the sum of all kinetic and potential energies of all components of a system

    - The change in internal energy $\Delta E = E_{final} - E_{initial}$

! When energy is exchanged between the system and the surroundings, it is exchanged as either heat or work

- $\Delta E = q + w$; q means the heat gained by the system, and w means the work done on the system

! The system absorbs heat from the surroundings → endothermic

! The system releases heat to the surroundings → exothermic

- State functions = quantities that are only dependent of the present state of the system and independent of the path by which the system achieved that state; e.g.: internal energy, temperature, volume…; q and w are not state functions

5.3 Enthalpy

- **$W = -P\Delta V$ (P constant)**

  => $\Delta E = q + w = q - P\Delta V$

- Enthalpy = internal energy plus the product of pressure and volume

- when the only work done is the pressure volume work:

$H = E + PV$; H means enthalpy, E means internal energy, P means pressure, and V means volume (all state functions)

- when the system changes at constant pressure:

$\Delta H = \Delta E + P\Delta V = (q_p + w) - w = q_p$

! $\Delta H > 0 \rightarrow$ endothermic; $\Delta H < 0$ exothermic

5.4 Enthalpies of Reaction

- **$\Delta H_{rxn} = H_{products} - H_{reactants}$; ΔH_{rxn} means the enthalpy of reaction or heat of reaction (q_{rxn})**

! Enthalpy is an extensive property

! ΔH for a reaction in the forward direction is equal in magnitude, but opposite in sign, to ΔH for the reverse reaction

! ΔH for a reaction depends on the states of the products and reactants

5.5 Calorimetry

- Calorimeter = a device that measures heat flow

ΔH measured through calorimetry

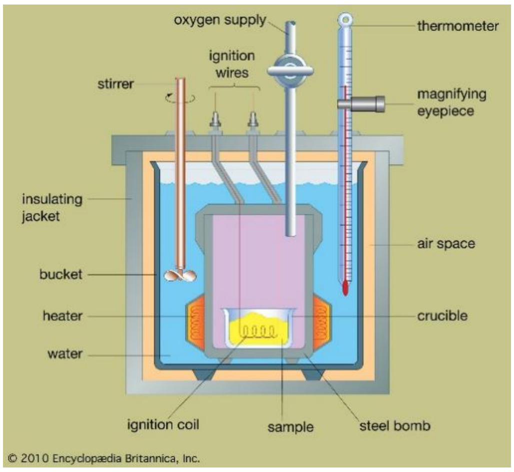
- Heat capacity (C) = the amount of energy required to raise the temperature of a substance by 1K (unit: JK^{-1})

- Specific heat capacity (C_s) = the amount of energy required to raise the temperature of 1 g of a substance by 1K (unit: $Jg^{-1}K^{-1}$)

- **$C_s = q/(m\Delta T)$; C_s means the specific heat, q means the heat transferred, m means the mass, and ΔT means the temperature change**

! For dilute aqueous solutions, we usually assume that the specific heat of the solution is the same as that of water, 4.18J/(g*K)

$q_{rxn} = - q_{soln} = - m_{soln} \times C_s \times \Delta T$

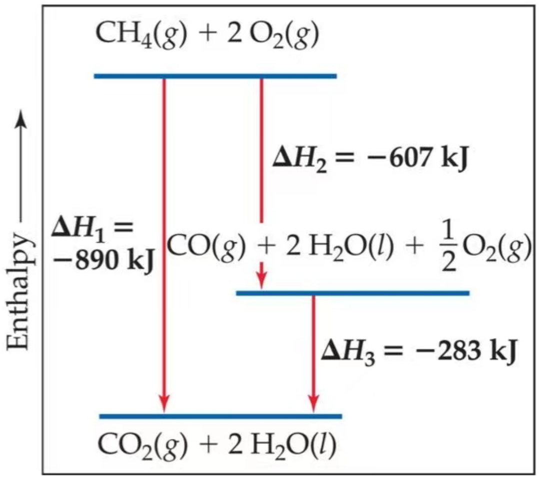


! Because the volume of the bomb calorimeter is constant, what is measured is really ΔE, not ΔH

q_{rxn} = - q_{cal} = - C_{cal} x ΔT; C_{cal} is the total heat capacity of the calorimeter

5.6 Hess's Law

- Hess's law = "if a reaction is carried out in a series of steps, ΔH for the overall reaction will be equal to the sum of the enthalpy changes for the individual steps"



5.7 Enthalpies of Formation

- Standard enthalpies of formation (ΔH^o_f) = the change in enthalpy for the reaction that forms one mole of the compound from its elements with all substances in their standard states

! ΔH^o_f of the most stable form of any element is zero

E.g. of stable forms: C(graphite), O_2, H_2

- $\Delta H^o_{rxn} = \sum(n\Delta H^o_f(\text{products})) - \sum(m\Delta H^o_f(\text{reactants}))$; n and m are stoichiometric coefficients

5.8 Strengths of Covalent Bonds

- Bond enthalpy = the enthalpy change, ΔH, for breaking of a particular bond in one mole of gaseous substance

! Average bond enthalpies are not absolute bond enthalpies

$\Delta H_{rxn} = \sum$(bond enthalpies of bonds broken) – $\sum$(bond enthalpies of bonds formed)

8.8 Strengths and Lengths of Covalent Bonds

! As the number of bonds between two atoms increases, the bond grows shorter and stronger, hence the bond enthalpy increases

(Chapter 13 is covered in the past resources)

GOOD LUCK ON YOUR FINALS :D